Lecture 2: Martingales and SDEs

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Outline

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Continuous Martingale and Stochastic Integrals

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Definition

Let $\mathcal{F}_n$ be an increasing sequence of σ-fields. $X_n \in \mathcal{F}_n$.
$X_n \in \mathcal{F}_n$ is a **submartingale (supermartingale)** if X_n is integrable for all $n \in \mathbb{N}$, and

$$\mathbf{E}(X_{n+1}|\mathcal{F}_n) \geqslant (\leqslant) X_n, \ a.s., \ \text{for all } n \in \mathbb{N}.$$

We say the filtration $\{\mathcal{F}_t\}_{t\in\mathbb{R}_+}$ satisfies the usual conditions if $\mathcal{F}_t$ is right continuous ($\mathcal{F}_t = \mathcal{F}_{t+} = \cap_{\varepsilon>0}\mathcal{F}_{t+\varepsilon}$). $X_t \in \mathcal{F}_t$ is a **submartingale (supermartingale)** if X_t is integrable for all $t \in \mathbb{R}_+$, and

$$\mathbf{E}(X_t|\mathcal{F}_s) \geqslant (\leqslant) X_s, \ a.s., \ \text{for all } s \leqslant t.$$

We say X_n (X_t) is a **martingale** if X_n (X_t) is a submartingale and supermartingale.

Optional stopping theorem

We say $T : \Omega \to \mathbb{N}\,(\mathbb{R}_+)$ is a stopping time if T satisfies

$$\{T \leqslant n\} \in \mathcal{F}_n \ (\{T \leqslant t\} \in \mathcal{F}_t).$$

We only state results for discrete-time martingales below. The following theorem lies at the basis of all other results for martingales.

Theorem 1 (Doob's Optional stopping theorem)

Assume that S is a bounded stopping time dominated by another bounded stopping time T, and X_n is a submartingale, then $\mathbf{E}(X_T|\mathcal{F}_S) \geqslant X_S$.

Doob's inequality

Theorem 2 (Doob's inequality)

Let M_n be a martingale. If $M_n^* := \sup_{k \leqslant n} |M_k|$, then

$$\mathbf{P}(M_n^* > \lambda) \leqslant \lambda^{-1} \mathbf{E}|M_n|.$$

Proof. Let $T = \inf\{k : |M_k| > \lambda\}$.

$$\begin{aligned} \lambda \mathbf{P}(M_n^* > \lambda) =& \lambda \mathbf{P}(T \leqslant n) \leqslant \mathbf{E}(|M_T|; T \leqslant n) \\ \leqslant& \mathbf{E}|M_{T \wedge n}| \leqslant \mathbf{E}|M_n|. \end{aligned}$$

Corollary 3

Let M_n be a martingale and T be a stopping time. For each $p > 1$, $\mathbf{E}|M_T^*|^p \leqslant C_p \mathbf{E}|M_T|^p$.

Martingale convergence theorem

$S_1 = \inf\{n : X_n \leqslant a\}$, $T_1 = \inf\{n > S_1 : X_n \geqslant b\}$,
$S_2 = \inf\{n > T_1 : M_n \leqslant a\}$, $T_2 = \inf\{n > S_2 : X_n \geqslant b\}$, ..., and
$U_n := \max\{k : T_k \leqslant n\}$.

Lemma 4 (Upcrossing inequality)

Suppose that X_n is a submartingale, then

$$(b-a)\mathbf{E}U_n(a,b) \leqslant \mathbf{E}(X_n - a)^+.$$

Sketch of Proof. We only prove the case that $a = 0$ and $X_k \geqslant 0$.

$$X_n = \underbrace{X_{S_1 \wedge n}}_{\geqslant 0} + \underbrace{\sum_{i=1}^{\infty} X_{T_i \wedge n} - X_{S_i \wedge n}}_{\geqslant bU_n(0,b)} + \sum_{i=1}^{\infty} \underbrace{X_{S_{i+1} \wedge n} - X_{T_i \wedge n}}_{\text{positive expectation}} .$$

Martingale convergence theorem

Upcrossing inequality leads to

Theorem 5

If X_n is a submartingale such that $\sup_n \mathbf{E} X_n^+ < \infty$, then X_n converges a.s. as $n \to \infty$.

Corollary 6

Suppose that $X \in L^1(\mathbf{P}, \Omega)$, $\mathcal{F}_n \uparrow \mathcal{F}_\infty$, then

$$\lim_{n\to\infty} \mathbf{E}(X|\mathcal{F}_n) = \mathbf{E}(X|\mathcal{F}_\infty), \quad \text{a.s. and in } L^1.$$

The above results also hold for right continuous submartingales (see [RY13]).

Maximal function

Let $\Omega = [0,1)^d$, $\mathbf{P}$ be Lebesgue measure, and f be an integrable function on Ω. Let $\mathcal{F}_n$ be the σ-field generated by the sets $\mathcal{A}_n = \{\Pi_{i=1}^d [k_i/2^n, (k_i+1)/2^n), k_i \in \mathbb{N}\}$. Let $f_n = \mathbf{E}(f|\mathcal{F}_n)$. Then f_n is a martingale, which is called **dyadic martingale**.

Corollary 3 and 6 implies that

Theorem 7

$$\|\sup_{n\geqslant 1} |f_n|\|_p \leqslant C_p \|f\|_p$$

and

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{1}{|Q_n(x)|} \int_{Q_n(x)} f(y)\mathrm{d}y = f(x), \quad a.e.$$

Exercise 1

Extend the Theorem 7 to the case that $f \in L^1(\mathbb{R}^d)$.

Calderón-Zygmund Decomposition

Martingale convergence theorem also provides the basis of the famous Calderón-Zygmund decomposition.

Lemma 8 (Calderón-Zygmund)

Let f be an integrable function on $\mathbb{R}^d$. Let $\lambda > 0$. There exists a sequence of pairwise disjoint cubes Q_i such that $f \leqslant \lambda$ on $(\cup_i Q_i)^c$, and

$$\lambda \leqslant \frac{1}{|Q_i|} \int_{Q_i} f \leqslant 2^d \lambda \quad , \quad \sum_i |Q_i| \leqslant \|f\|_1 / \lambda.$$

The reader is referred to [Bas94] for more applications of probabilistic techniques.

Continuous Martingales

Theorem 9 (Doob-Meyer decomposition)

Assume X is a continuous submartingale, then there exists a unique martingale M and a unique continuous increasing adapted process A such that

$$A_0 = 0, \quad X_t = M_t + A_t.$$

If M is a continuous square integrable martingale, then M^2 is a submartingale. Thus, there exists a continuous increasing process, denoted by $\langle M \rangle$, the **quadratic variation** of M, such that $M^2 - \langle M \rangle$ is a martingale. Particularly, $\mathbf{E}M_t^2 - \mathbf{E}M_0^2 = \mathbf{E}\langle M \rangle_t$.

Construction of stochastic Integrals

Let M be a square integrable martingale. Let

$$\mathcal{S} = \left\{ H : H_t = \sum_{i=1}^{N} H_{t_i} \mathbf{1}_{[t_i, t_{i+1})}(t),\ H_{t_i} \in \mathcal{F}_{t_i} \text{ and } H_{t_i} \text{ is bounded} \right\}.$$

For simple process $H \in \mathcal{S}$, define

$$(H \cdot M)_t = \int_0^t H_s \mathrm{d}M_s = \sum_i H_{t_i}(M_{t_{i+1} \wedge t} - M_{t_i \wedge t}).$$

Then it is not hard to show that $H \cdot M$ is a continuous martingale and

$$\langle H \cdot M \rangle_t = \int_0^t H_s^2 \mathrm{d}\langle M \rangle_s. \tag{1}$$

Construction of stochastic Integrals

If H can be approximated by $H^n \in \mathcal{S}$ in the sense that

$$\mathbf{E} \int_0^\infty (H_s^n - H_s)^2 \, \mathrm{d}\langle M \rangle_s \to 0, \tag{2}$$

then there exists a unique continuous martingale X such that

$$\mathbf{E} \left(\sup_{t \geqslant 0} |(H^n \cdot M)_t - X_t|^2 \right) \leqslant C \mathbf{E} \int_0^\infty (H_s^n - H_s)^2 \, \mathrm{d}\langle M \rangle_s \to 0.$$

We denote X by $H \cdot M$.

Extension

Let $\mathcal{P}$ be the σ-field on $\Omega \times \mathbb{R}_+$ generated by $\mathcal{S}$.

Exercise 2

Assume that H is a $\mathcal{P}$-measurable function satisfies $\int_0^\infty H_s^2 \, \mathrm{d}\langle M\rangle_s < \infty$, then there exists a sequence $H^n \in \mathcal{S}$ such that (2) holds.

For square integrable martingale M, using stopping time techniques, one can define $H \cdot M$ for all H satisfying $\int_0^t H_s \, \mathrm{d}\langle M\rangle_s < \infty$ $(\forall t \geqslant 0)$. Similarly, the integration $H \cdot M$ can be extended to the case that M is a **local martingale**. If X is a **semimartingale** with decomposition $X = M + V$, and M is a local martingale and $\int_0^t \mathrm{d}|V|_s < \infty$, then for all $H \in \mathcal{P}$ satisfying $\int_0^t H_s^2 \mathrm{d}\langle M\rangle_s + \int_0^t |H_s| \, \mathrm{d}|V|_s < \infty$ $(\forall t > 0)$, we define

$$(H \cdot X)_t = (H \cdot M)_t + \int_0^t H_s \, \mathrm{d}V_s.$$

Itô's formula and its applications

The most important result about stocahstic integration is

Theorem 10 (Itô's formula)

Let X be a continuous semimartingale. Let $f \in C^2$, then

$$f(X_t) - f(X_0) = \int_0^t f'(X_s)\,\mathrm{d}X_s + \int_0^t \frac{1}{2} f''(X_s)\,\mathrm{d}\langle X\rangle_s. \tag{3}$$

Itô's formula can be used to prove the following

Theorem 11 (BDG inequality)

Suppose M is a local martingale with continuous paths and $M_0 = 0$. Assume that T is a stopping time and $p > 0$, then there exist universal constants c_p and C_p such that

$$c_p \mathbf{E}\langle M\rangle_T^{\frac{p}{2}} \leqslant \mathbf{E}(M_T^*)^p \leqslant C_p \mathbf{E}\langle M\rangle_T^{\frac{p}{2}}. \tag{4}$$

Itô's formula and its applications

Sketch of Proof.

- For any $p \geqslant 2$, by Itô's formula

$$|M_T|^p = p\int_0^T \mathrm{sgn}(M_t)|M_t|^{p-2}M_t\mathrm{d}M_t + \frac{p(p-1)}{2}\int_0^T |M_t|^{p-2}\mathrm{d}\langle M\rangle_t;$$

- By Doob's inequality and Hölder's inequality,

$$\begin{aligned}\mathbf{E}(M_T^*)^p \lesssim_p \mathbf{E}|M_T|^p \lesssim_p \mathbf{E}((M_T^*)^{p-2}\langle M\rangle_T)\\ \leqslant (\mathbf{E}(M_T^*)^p)^{1-\frac{2}{p}}(\mathbf{E}\langle M\rangle_T^{\frac{p}{2}})^{\frac{2}{p}};\end{aligned}$$

- Using Lenglart's domination inequality to get the case $p \in (0, 2)$.

Another approach to proving (4) is utilizing "good-λ" inequality (cf. [RY13]).

Itô's Equation

Given a and b, we want to construct a diffusion process X with generator $\mathcal{L} = \frac{1}{2} a_{ij} \partial_{ij} + b_i \partial_i$.

Itô's Equation

Given a and b, we want to construct a diffusion process X with generator $\mathcal{L} = \frac{1}{2} a_{ij} \partial_{ij} + b_i \partial_i$.

Let W be a Brownian motion on a filtered probability space $\mathbb{F}$ (satisfies the usual conditions). For each $\xi \in \mathcal{F}_0$, consider the stochastic differential equation (SDE)

$$X_t^\xi = \xi + \int_0^t \sigma(X_s^\xi) \, \mathrm{d}W_s + \int_0^t b(X_s^\xi) \, \mathrm{d}s. \tag{5}$$

Itô's Equation

Since we will talk about various types of solutions later, to be precise, we say that X is a **strong solution** if X is adapted to the filtration generated by W, and satisfies (5). We say that we have pathwise uniqueness for (5) if whenever X and Y are two strong solutions, then

$$\mathbf{P}(X_t = Y_t, \text{ for all } t \geqslant 0) = 1.$$

Theorem 12

Assume W is a n-dimensional Brownian motion. Suppose that $\sigma : \mathbb{R}^d \to \mathbb{R}^{d\times n}$ and $b : \mathbb{R}^d \to \mathbb{R}^d$ both are locally Lipschitz functions with linear growth, then for each $\xi \in L^2(\Omega, \mathcal{F}_0; \mathbf{P})$, there is a strong solution to (5)*, and the solution is pathwisely unique.*

Itô's Equation

Sketch of Proof. We assume σ and b are bounded and Lipschitz. Let B be the collection of all continuous semimartingale on $[0, t_0]$ equipped with norm

$$\|X\|_B := [\mathbf{E}(X_{t_0}^*)^2]^{1/2}.$$

Define $T : B \to B$ as

$$[T(X)]_t = \xi + \int_0^t \sigma(X_s)\,\mathrm{d}W_s + \int_0^t b(X_s)\,\mathrm{d}s, \quad t \in [0, t_0].$$

Doob's inequality and BDG inequality (or Itô's isometry) yield

$$\|T(X^1) - T(X^2)\|_B \leqslant C\sqrt{t_0}\|X^1 - X^2\|_B \leqslant \frac{1}{2}\|X^1 - X^2\|_B,$$

provided that t_0 is small. Thanks to Banach's fixed point theorem, T has a unique fixed point.

Markov properties

Although we have constructed a family of stochastic processes X, we do not know whether the constructed processes are Markovian. It turns out that if pathwise uniqueness holds for (5) for every x, then X is Markov. However, it is no more difficult to prove that a weaker type of uniqueness also implies the Markov property.

References

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